THE INTERNAL WAR

Could gut cells, microbes and the brain work together to steady glucose and hunger?

The discovery and the question

A May 2026 study found that metformin and berberine can inhibit mitochondrial Complex I in intestinal cells. Those cells compensate by using more glucose, creating a glucose sink [1]. Could gut microbes and brain appetite signals be linked to this switch?

Why slowing one component increases glucose use

Complex I normally helps mitochondria generate ATP using fuels from both glucose and fat. It is not an inhibitor of glucose burning. Restricting it pushes intestinal cells towards glycolysis, which uses more glucose for a smaller energy return.

L-lactate, Lac-Phe and a brake on a brake

Glycolysis makes pyruvate, some of which becomes L-lactate. CNDP2 links lactate to the amino acid phenylalanine to make Lac-Phe. In mice Lac-Phe quietens AgRP hunger neurons, reducing their inhibition of satiety circuits [2-4].

The Ford Cortina test

Imagine a Mark II with its choke stuck out. Once the settings are corrected, it works properly. Complex I is not exactly a faulty choke; turning it down has costs and risks. A working engine needs the correct controls.

The glucose-sink pathway

Complex I limited > More glycolysis > Pyruvate > L-lactate

L-lactate + phenylalanine --CNDP2--> Lac-Phe --| hunger neurons

FROM ASGARD TO APPETITE

An ancient partnership, today’s microbiome, and a hunger signal.

Where mitochondria came from

The host was an archaeon related to Asgard archaea, especially Heimdallarchaeia. New Hodarchaeales relatives were cultured in 2026 [5]. A bacterium related to Alphaproteobacteria became our mitochondrial ancestor. Microbes already made ATP by fermentation and membrane respiration.

Gut bacteria work as teams

Fermentable fibre feeds microbes that produce acetate (C2), propionate (C3) and butyrate (C4). Some bifidobacteria feed other microbes that make butyrate. Colon cells use butyrate as a healthy fuel. The liver can use microbial propionate to help make odd-chain fatty acids, including C15 [6-8].

Yoghurt and lactate

Some yoghurt, kefir and cheese contain L-lactate, D-lactate or mixtures, depending on fermentation. A human experiment found oral lactate raised L-Lac-Phe [9,10]. Whether an ordinary fermented-food snack changes evening cravings is an appealing research question.

Vinegar, HIIT and the evening

Apple cider vinegar can dampen some glucose rises, but may worsen slow gastric emptying [11]. Exercise, especially high-intensity interval training (HIIT), raises lactate and Lac-Phe [12]. Ordinary movement helps too. Evening cravings also follow a biological clock [13].

Our question

Can well-tolerated food-derived or microbial signals improve intestinal glucose disposal while preserving healthy butyrate use and satiety signalling?

An ancient partnership

Asgard archaeal host + Alphaproteobacterium > Mitochondrion

SOURCES & YOUR CHALLENGE

Research to test, question and improve. Click any source below.

THE SOURCES | Click each citation to open the research

01. Sebo et al. 2026. Intestinal Complex I and glucose sink

02. Xiao et al. 2024. Metformin, CNDP2 and Lac-Phe

03. Scott et al. 2024. Lac-Phe after metformin and meals

04. Liu et al. 2025. Lac-Phe and AgRP hunger neurons

05. Imachi et al. 2026. Cultured Heimdallarchaeia

06. Culp and Goodman 2023. Microbial cross-feeding

07. Byndloss et al. 2017. Colonocytes and butyrate

08. Weitkunat et al. 2017. Inulin, propionate and C15

09. Ge et al. 2024. Yoghurt L- and D-lactate

10. Human oral lactate study 2025. L-Lac-Phe

11. Hlebowicz et al. 2007. Vinegar and gastric emptying

12. Li et al. 2022. Exercise-induced Lac-Phe

13. Scheer et al. 2013. Circadian appetite

What should we investigate next?

Could microbial metabolites change the fuel choices of intestinal cells? Could fermented foods alter Lac-Phe or evening hunger? How do we separate glucose disposal from appetite effects?

LEAVE A DOT: Challenge, correct or extend this hypothesis

github.com/KarstenEvans/aletheia/discussions